

PART-B

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INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	4	Total
Points:	6	7	5	2	20
Score:	0	7	5	2	14

QUESTIONS

1. (6 points) Let

$$S = \begin{pmatrix} S_{yy} & S_{yX}^T \\ S_{yX} & S_{XX} \end{pmatrix}$$

be the partitioned sample covariance matrix of $Y_{1 \times 1}$ and $X_{q \times 1}$ based on a centered (with respect to sample mean) sample of size n . Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector X .

$$R^2 = \frac{SS_{\text{Reg}}}{SS_{\text{Tot}}} = \frac{\sum_{i=1}^n (\bar{y}_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = \frac{\sum_{i=1}^n (\hat{y}_i)^2}{\sum_{i=1}^n y_i^2}$$

For ~~max~~ first canonical correlation, we need

$$\max_{a, b} \text{corr}(a^T y, b^T X) \quad \text{such that} \quad \text{Var}(a^T y) = 1 \quad \text{and} \quad \text{Var}(b^T X) = 1$$

But since y is dimensional, a^T is a constant.

Since multiplying by a constant doesn't change correlation,

we can ignore a^T .

So we want $\max_b \text{corr}(y, b^T X)$

$$\max_b \frac{\text{cov}(y, b^T X)}{\sqrt{\text{Var}(y) \text{Var}(b^T X)}}$$

$$\max_b \frac{\text{cov}(y, b^T X)}{\sqrt{\text{Var}(y)}}$$

$$\max_b \frac{\sum_{i=1}^n y_i (b_1 x_{i1} + b_2 x_{i2} \dots b_n x_{in})}{\sqrt{\sum_{i=1}^n y_i^2}}$$

$$2 \quad \max_b \frac{\hat{y}^T \hat{y}}{\sum_{i=1}^n y_i^2}$$

$$3 \quad \max_b \frac{(y - \hat{y} + \hat{y})^T \hat{y}}{\sum y_i^2}$$

$$2 \quad \max_b \frac{\hat{y}^T \hat{y} + (y - \hat{y})^T \hat{y}}{\sum y_i^2}$$

• If b is the value of weights such that ~~we minimize~~

$(y - \hat{y})^T (y - \hat{y})$ is minimized, then $(y - \hat{y})^T \hat{y} = e^T \hat{y} = 0$
(Proof is below)

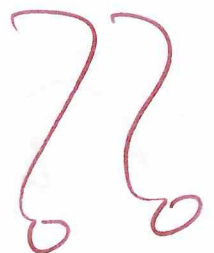
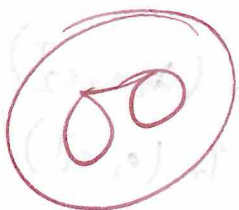
$$= \frac{\sum_i (\hat{y}_i)^2}{\sum y_i^2} = r^2$$

We can see that $r^2 = R^2$ How?!

Maximizing $y^T \hat{y}$ is equivalent to minimizing $(y - \hat{y})^T (y - \hat{y})$

$$\begin{aligned} \text{Since } (y - \hat{y})^T (y - \hat{y}) &= y^T y - 2\hat{y}^T \hat{y} + \hat{y}^T \hat{y} \\ &= 2(1 - 2y^T \hat{y}) \end{aligned}$$

(Since y and $\hat{y}$ are normalized).



2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix $\mathbf{S}$. The eigenvalues of $\mathbf{S}$ are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and $\mathbf{e}_4$. You decide to extract $m = 2$ factors using the principal component method.
- (a) (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{\mathbf{L}}$ in terms of the given eigenvalues and eigenvectors.
- (b) (1 point) Let the first row of $\tilde{\mathbf{L}}$ be $\tilde{\mathbf{l}}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.
- (c) (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^p \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of $\mathbf{S}$.

(a) $\tilde{\mathbf{L}} = \left[\sqrt{4.2} \mathbf{e}_3 \mid \sqrt{2.1} \mathbf{e}_2 \right]$

(b) $\tilde{\mathbf{L}} = \begin{bmatrix} 0.8 & -0.4 \\ \vdots & \vdots \end{bmatrix}$

For the first variable, communality is given as

(b) $\sum_{i=1}^2 \lambda_{1i}^2 = (0.8)^2 + (-0.4)^2 = 0.8$

(c) $\tilde{\mathbf{L}} = \left[\sqrt{\lambda_3} \mathbf{e}_3 \mid \sqrt{\lambda_2} \mathbf{e}_2 \right]$ since λ_3 and λ_2 are the two largest eigenvalues

$$\tilde{\mathbf{L}} \tilde{\mathbf{L}}^T = \begin{bmatrix} \sqrt{\lambda_3} \mathbf{e}_3 & \sqrt{\lambda_2} \mathbf{e}_2 \end{bmatrix} \begin{bmatrix} \sqrt{\lambda_3} \mathbf{e}_3^T \\ \sqrt{\lambda_2} \mathbf{e}_2^T \end{bmatrix} = \lambda_3 \mathbf{e}_3 \mathbf{e}_3^T + \lambda_2 \mathbf{e}_2 \mathbf{e}_2^T$$

By definition, $\sum_{i=1}^p \tilde{h}_i^2 = \text{Tr}(\tilde{\mathbf{L}} \tilde{\mathbf{L}}^T)$

$$= \text{Tr}(\lambda_3 \mathbf{e}_3 \mathbf{e}_3^T) + \text{Tr}(\lambda_2 \mathbf{e}_2 \mathbf{e}_2^T)$$

$$= \lambda_3 \text{Tr}(\mathbf{e}_3 \mathbf{e}_3^T) + \lambda_2 \text{Tr}(\mathbf{e}_2 \mathbf{e}_2^T)$$

Now by property of Transpose, $\text{Tr}(\mathbf{AB}) = \text{Tr}(\mathbf{BA})$

Also since $\mathbf{e}_3$ and $\mathbf{e}_2$ are normalized eigen vectors,

$$e_3^T e_3 = 1 \quad \text{and} \quad e_2^T e_2 = 1$$

We have,
$$\sum_{i=1}^p h_i = \lambda_3 \operatorname{Tr}(e_3^T e_3) + \lambda_2 \operatorname{Tr}(e_2^T e_2)$$

$$= \lambda_3 + \lambda_2$$

which is exactly the sum of largest two eigenvalues of S .

3. (5 points) Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2 . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Derive a variance-stabilizing transformation $g(\bar{X}_n)$ such that the asymptotic variance of $g(\bar{X}_n)$ does not depend on μ .

Ans.) $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$

From CLT, $\sqrt{n} \frac{(\bar{X}_n - \mu)}{\mu} \xrightarrow{d} N(0, 1)$ as $n \rightarrow \infty$

From Taylor's thm,

$$g(\bar{X}_n) \approx g(\mu) + g'(\mu')(\bar{X}_n - \mu)$$

where μ' lies in $(\min(\bar{X}_n, \mu), \max(\bar{X}_n, \mu))$

$$\Rightarrow \sqrt{n}(g(\bar{X}_n) - g(\mu)) = g'(\mu') \sqrt{n}(\bar{X}_n - \mu)$$

As $n \rightarrow \infty$, $\mu' \xrightarrow{d} \mu$

~~Assume~~ Assume ~~g'~~ g' is a continuous function $g'(\mu') \rightarrow g'(\mu)$.

~~Also~~ Also $\sqrt{}$

So, as $n \rightarrow \infty$,

$$\sqrt{n}(g(\bar{X}_n) - g(\mu)) \xrightarrow{d} g'(\mu) \mu N(0, 1)$$

~~the~~ If the asymptotic variance is to remain independent of μ ,

$$g'(\mu) \mu = c_1$$

$$\Rightarrow \frac{dg(\mu)}{d\mu} = \frac{c_1}{\mu}$$

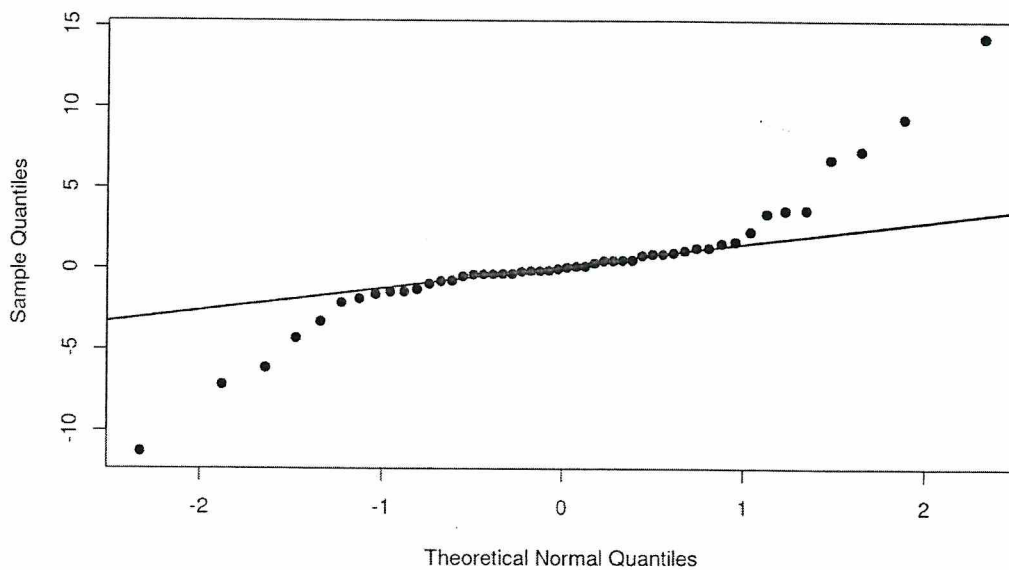
OS $\Rightarrow \boxed{g(\mu) = c_1 \ln \mu + c_2}$

So $g(\mu) = c_1 \ln \mu + c_2$ is a variance stabilizing transformation.

For $c_1 = 1$, $c_2 = 0$, we get

$$\boxed{g(\mu) = \ln \mu}$$

4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



No, the data doesn't follow normal distribution since for data to follow Normal distribution, all points should approximately lie on a single line.

Since the data is approximately normal in the middle but diverges as the points get extreme, we should follow a more heavy-tailed distribution like Cauchy.

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ADDITIONAL PAGES FOR ANSWERS
